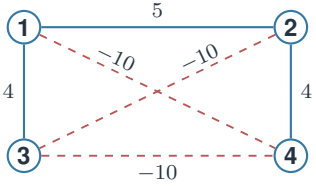
(a) Merge rule: signed gain and structural feasibility



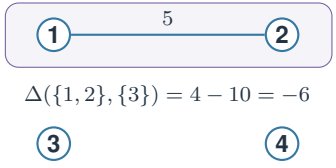
After a merge: add gains, AND feasibility. Feasibility is guaranteed; global objective optimality is not.

(b) Four-node counterexample



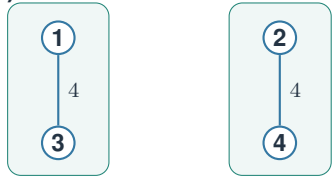
All receivers distinct; all gates admit.

(c) Greedy result: $J = 5$



No remaining positive feasible merge.

(d) Exact result: $J = 8$



Early greedy merge must be split.